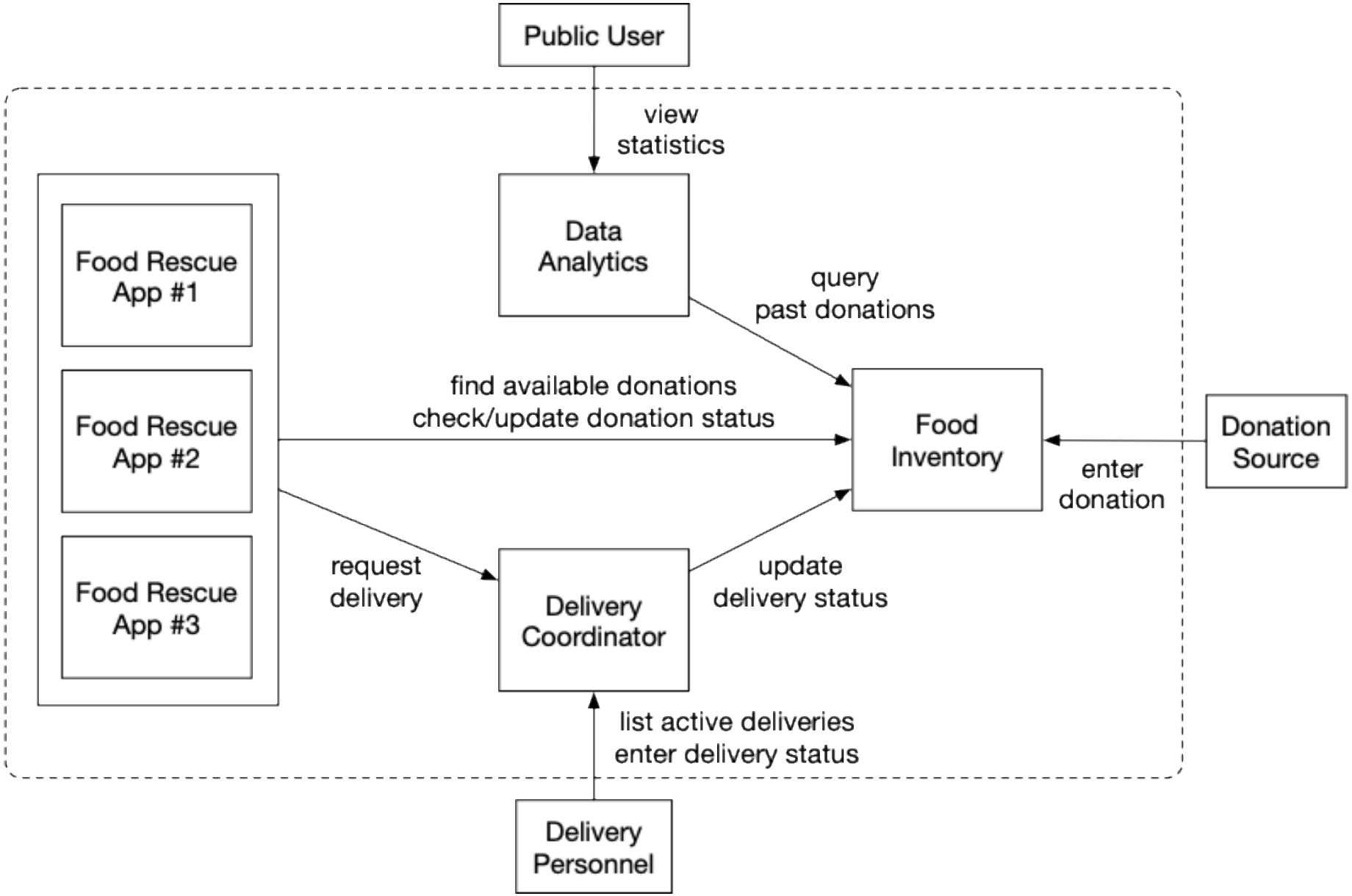
17-423/723: Software System Design

Interface Design Session

Mar 9, 2026

Today’s Objectives

- Over the next project milestones (M3 and M4), your team will design and implement a shared service that is used by multiple teams
- In today’s class, your team will:
  - Begin designing an interface for your shared service
  - Share your design and provide feedback on others
  - Refine the interface design based on the feedback


Public User
view
statistics
Data
Analytics
query
past donations
Food Rescue
App #1
Food Rescue
App #2
Food Rescue
App #3
find available donations
check/update donation status
Food
Inventory
Donation
Source
enter
donation
request
delivery
Delivery
Coordinator
update
delivery status
list active deliveries
enter delivery status
Delivery
Personnel

Design Session Plan

1. Identify use cases (10 min)
2. Design an initial version of API (15 min)
3. Share the design (10 min)
4. Review & critique (15 min)
5. Clarify & refine (10 min)

Identify use cases (10 min)

- **Goal**: Understand how this service will be used by and interact with the other services & food rescue apps
- Brainstorm & identify use case scenarios that involve your team's service
  - Example: "A recipient claims a donation and requests delivery"
  - Think about not just normal successful cases, but also edge cases (e.g., what if no donation data available for a given time period?)
  - Sketch these scenarios out using sequence diagrams!
- Identify functions that your service needs to provide to support these use cases

Design an initial version of API (15 min)

- For each API function, identify:
  - **A list of input & output parameters**
  - **Contract** (i.e., pre- & post-conditions)
  - **Data ontology**, describing for each parameter:
    - Type (syntax) and meaning (semantics) of the parameter
    - Any constraints over the parameter
    - Any relationships with other parameters
- Document your design in the shared Google Doc
- Nominate one person to present the design to the class

Share your design (10 min)

- Briefly describe the API for your team’s service to the class

Review & Critique (15 min)

- Read and discuss the APIs for the other teams' services on the shared document
- Consider:
  - Does the API support the use case scenarios identified in Step #1?
  - Is the contract for each API well-defined? Is it general enough or too restrictive? Is it too strong or too weak?
  - Are the parameters well-specified?
  - Does it describe what happens under edge/failure cases?
- Leave questions or suggestions for changes directly using Google Doc comments

Clarify & Refine (10 min)

- Read the comments on your team's API
- Ask any clarifying questions
- Modify and refine the API based on the comments

Next Steps

- Finalize the APIs by **Friday, March 13**
  - Other teams will depend on your API being stable; if you make any changes to it, it might impact their code as well!
- Implement the new feature (i.e., delivery request) for your food rescue app
  - Since the Delivery Coordinator will not be available yet, create a test double for testing your feature
- In Milestone 4, you will implement your shared service and integrate it with the other teams